Autrin Hakimi

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PROFESSIONAL PROFILE

Software engineer building **full-stack applications**, **ML-driven systems**, and **robotics automation**. Currently at ASM, where I independently own and deliver multiple full-stack systems for semiconductor manufacturing. Published researcher (**ACM SIGSPATIAL 2025**) with experience in distributed deep learning across **64 NVIDIA A100 GPUs**.

TECHNICAL SKILLS

Languages Python, C/C++, JavaScript, TypeScript, SQL, Bash, HTML, CSS
Web & Backend React, Python Flask, Tkinter, REST APIs, WebSockets, JavaScript/HTML/CSS
Cloud & DevOps Docker, Git, CI/CD pipelines, PyInstaller/NSIS packaging, Linux/Unix, Microsoft Azure, Google Cloud, Snowflake
ML & Data PyTorch, scikit-learn, Pandas, NumPy, distributed training (64-GPU), DGL, NetworkX
Databases MySQL, SQL, SQLite, NoSQL, Snowflake
Automation & Control Robot workflow orchestration, vision systems, HMI/GUI, ROS, OpenCV, Yamaha robot controllers
Methodologies Agile (Scrum/Kanban), test-driven development, cross-functional collaboration

EDUCATION

B.S., Computer Science **Aug 2021 - May 2025**
Iowa State University | GPA: 3.70/4.0 | Magna Cum Laude | Dean's List *Ames, IA*

PROFESSIONAL EXPERIENCE

Software Engineer **Jan 2026 - Present**
ASM America *Phoenix, AZ*

- Independently own and deliver multiple **full-stack production systems** for semiconductor manufacturing: robotics platforms, an ML-driven parts matching tool, and operator tooling.
- Authored the majority of the codebase (backend and frontend) for the operator-facing **collaborative robot application** shipping on a **seven-figure robotics product**: architected the **HMI and operator controls** (guided workflows, manual jog, abort), engineered **multithreading** so live camera feeds and robot motion never freeze the UI, and built **recovery and motion-safety logic** including motion guards, safe return-to-home, network retry/reconnect, and FTP file transfer; a **major customer's multi-unit order worth tens of millions** ships on this product, and the UI rebuild earned a **quarterly team award**.
- Improved measurement accuracy and run reliability on a **robot/camera conductance-measurement fixture**, correcting motion, calibration, and geometry logic across many part configurations; built **automated validation suites (140+ tests)** verifying bore-profile geometry and control logic off-hardware.
- Built a full-stack **ML-driven parts matching system** (Flask, JavaScript, SQL) that saves engineers hours of manual lookup, now the basis for a **pending patent** on which I am a named inventor; engineered **threading and synchronization** across robotics frontends to coordinate vision and control workflows.
- Package and deploy applications to international manufacturing sites with PyInstaller and NSIS, and provide remote troubleshooting; this accelerated release cycles and lowered field-support response time
- Lead **requirements definition and stakeholder alignment** to digitize a customer-facing qualification workflow, introducing Git version control for the team, which improved code traceability and reduced integration errors

Undergraduate AI Researcher **Aug 2024 - Nov 2025**
Iowa State University, SwAPP Lab *Remote*

- Co-authored “HydroGAT” [paper, code] accepted at **ACM SIGSPATIAL 2025**: a heterogeneous graph attention transformer for flood prediction reaching **NSE 0.97**; built distributed training across **64 NVIDIA A100 GPUs** on NERSC Perlmutter with **15× speedup**.
- Engineered end-to-end **data pipelines** (PyTorch, DGL, Pandas, NumPy) for large-scale datasets; automated workflows via Bash scripting, cutting setup time **40%**.

Software Development Intern **May 2024 - Aug 2024**
BuilderTREND *Omaha, NE*

- Designed and shipped a **React**-based job proposal template; migrated legacy ASP pages to React and built full-stack features in JavaScript, TypeScript, and C# (.NET) within Kanban/Scrum sprints.

Data Science Intern **May 2023 - May 2024**
Alliant Energy *Remote*

- Developed **predictive ML models** (Random Forest, SVM) in Python and SQL analyzing electricity usage for **995K customers** in Snowflake; predicted EV ownership for customer segmentation, recognized by the Director of Data Analytics for impactful business insights.

SELECTED PROJECTS

Defect Sense: Industrial Visual Inspection [code]

- Built a **two-stage industrial defect-detection pipeline**: a fast anomaly detector (**PatchCore/EfficientAD**, Anomalib) routes uncertain parts to a **local vision-language model** (Qwen3-VL via Ollama) for defect typing and reporting; validated the detector across all **15 MVTec AD categories (0.982 image AUROC)**, served via a **FastAPI** console with schema-constrained output, GPU-free CI.

Autonomous Robotic Chess System [code]

- Built a fully autonomous system: piece recognition (**OpenCV**), move calculation, and **UR10e robot arm control** (ROS, MoveIt).